Harshana Lakshara Fernando

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SUMMARY

Full-Stack Software Engineer experienced in end-to-end application delivery, legacy integrations, and production support. Strong in **Java**, **Spring Boot**, **Kafka**, **React**, and DevOps workflows. Known for rapid learning, taking ownership of distributed systems, and choosing solutions that balance reliability, scalability, and real product needs.

EDUCATION

B.Sc. (Hons) in Engineering — Computer Science and Engineering, University of Moratuwa 2022 – 2026
G.C.E. A/L: 3 A's, Physical Science Stream, District Rank 17

EXPERIENCE

Software Engineer — Trading Systems Sep 2026 – Present
CAL Colombo, Sri Lanka

- Building CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus dealing, operations and testing tools.
- Led CSE's **188**-case **exchange certification** with a Certification Runner that drives each case and diffs every field, backed by message logs and a CI-enforced register.
- **Event-sourced CQRS** order path on a **Chronicle Queue** journal with **PostgreSQL** as async read model; **FIX 5.0 SP2** routing and drop copy (**QuickFIX/J**) and a **Java NIO MITCH** multicast decoder.
- Built an exchange simulator with **zero parity failures** over **1,700+** real exchange conversations, failover and gap recovery, and **~1,050 automated tests** in CI.

R&D Software Engineer — KōreroCare Platform Mar 2026 – Aug 2026
Empathic Computing Laboratory, University of Auckland Remote

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.

Co-founder & Lead Engineer (Freelance) Jul 2025 – Aug 2026
ScaleX — SaaS Development Lab Remote / Sri Lanka

- Co-founded ScaleX and owned the end-to-end lifecycle of multiple SaaS MVPs, from architecture through cloud-native deployment.
- Built **AgriNova** (**Next.js**, **Prisma**, **Supabase**/**PostgreSQL**), an inventory and commission platform across 5 branches and 329+ SKUs, tracking 194K+ units and 860+ invoices (Rs. 64.4M+).
- Cut repeat dashboard data-fetch latency from ∼384ms to <1ms with **React Query** caching and pagination.

Software Engineer Intern — ISA Aviation Technology Dec 2024 – May 2026
Information Systems Associates (Pvt) Ltd Colombo, Sri Lanka

- Owned design and delivery of a distributed **Spring Boot**/**Kafka**-backed PNR ID generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
- Designed a **pool-based PNR allocation model** replacing legacy Oracle-sequence generation, with **concurrency-safe** Kafka-backed allocation and Java **ReentrantLock** critical sections.
- Led the rollout to AeroOrder using database-driven **feature toggles** and a **strangler-pattern** migration, bridging the **Java 8** AeroMart monolith via **gRPC** clients and Kafka consumers.
- Resolved production **Kafka consumer lag** and offset-reset issues with graceful shutdown hooks, retry handling, validation, and a Producer–Consumer refactor.
- Built load testing and monitoring with **JMeter**, **Prometheus**, and **Grafana**; maintained **>90% unit test coverage**.

OPEN SOURCE

Merged: KEDA #8317 (CNCF, `onlyReadyPods` scaler option, docs and a follow-up refactor); kgateway #14851 (CNCF) — `kgateway_build_info` metric, shipped in v2.5.0-beta.2; kubernetes-mcp-server #1482 and #1549 — test-startup fix, RBAC metadata test.
In review: BanyanDB #1462 (Apache, MCP Bearer-auth fix for a bug I found); Jaeger #9997 (CNCF, MCP topology fix for a bug I reported); kgateway #14945 (local rate-limit `statPrefix`); Spark Operator #3249 (executor-metrics tests, LGTM).

PROJECTS

Supply Chain DApp Solidity, Next.js, Polygon

- Developed a full-stack supply chain tracker with trustless escrow.
- Technologies: Solidity, Next.js

Governance DAO Solidity, Hardhat

- Engineered a decentralized autonomous organization for voting.
- Technologies: Solidity, OpenZeppelin

ZK-Rollups Scalability Circom, Solidity

- Researched batching and throughput enhancement on Ethereum Layer 2.
- Technologies: Circom, Solidity

Fintech & Crypto Lab. Research

- Group project focusing on decentralized finance mechanisms.
- Technologies: DeFi, Blockchain

SKILLS

Backend: Java, **Spring Boot**, Go/Gin, Python, Node.js, REST, **gRPC**, GraphQL, Hibernate
Frontend: React, Next.js, **TypeScript**, JavaScript, React Query
Messaging & APIs: Kafka, RabbitMQ, WebSockets, Protobuf, JWT, OAuth2/OIDC
Data: PostgreSQL, MySQL, Oracle DB, Redis, Couchbase, DynamoDB, Prisma
Cloud / DevOps: AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux
Observability / Testing: Prometheus, Grafana, JMeter, Gatling, JUnit, Mockito, Pytest